Hints: Methods of proof – 1

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Note. The last claim is false with the fractional part $\{q\alpha\}$: for $\alpha = 1 - \varepsilon$ with small $\varepsilon > 0$ one has $\{q\alpha\} = 1 - q\varepsilon > 1/N$ for all $q \leqslant N$. The correct statement uses the distance to the nearest integer: there is $q \leqslant N$ with $\|q\alpha\| < 1/N$. The hints prove this version, which is enough for the density.

Hint 1. For the pigeonhole: if every box held at most $\lceil n/k \rceil - 1$ objects, the total would be at most $k(\lceil n/k \rceil - 1) < n$. For the approximation, look at the $N+1$ numbers $\{0 \cdot \alpha\}, \{\alpha\}, \ldots, \{N\alpha\}$ and the N boxes $[j/N, (j+1)/N)$, $j = 0, \ldots, N-1$.

Hint 2. Two of them, $\{a\alpha\}$ and $\{b\alpha\}$ with $a < b$, share a box. With $q = b - a \leqslant N$ this gives an integer p with $|q\alpha - p| < 1/N$, so $\|q\alpha\| < 1/N$, and then $|\alpha - p/q| < 1/(qN) \leqslant 1/q^2$. To get infinitely many fractions: since α is irrational, $|\alpha - p/q| > 0$, so choose N with $1/N$ smaller than this error and get a new fraction.

Hint 3. Density: given an interval of length ε in $[0,1]$, take $N > 1/\varepsilon$ and q with $\delta = \|q\alpha\| < 1/N$, $\delta > 0$. The points $\{mq\alpha\}$, $m = 1, 2, \ldots$, then move along the circle $\mathbb{R}/\mathbb{Z}$ in steps of exactly δ (forwards if $\{q\alpha\} = \delta$, backwards if $\{q\alpha\} = 1 - \delta$). Steps shorter than ε cannot jump over the interval.

Problem 2

Hint 1. Blichfeldt: integrate $\sum_i \mathbf{1}_{A_i}$ over the ambient set of measure μ. If every point lay in at most k of the sets, the integral would be at most $k\mu$. In the lattice version, translate a set $S \subseteq \mathbb{R}^n$ of volume > 1 into the unit cube $[0,1)^n$ piece by piece (one piece for each integer translate). The pieces have total volume > 1, so two points $x \neq y$ of S have $x - y \in \mathbb{Z}^n$.

Hint 2. Minkowski: apply this to $S = \frac{1}{2}K$, which has volume > 1 when $\operatorname{vol} K > 2^n$. Then $x - y = \frac{1}{2}(2x) + \frac{1}{2}(-2y) \in K$ by symmetry and convexity, and $x - y$ is a non-zero lattice point.

Hint 3. Steinhaus: we may shrink A to a bounded set of positive measure inside an interval I. For $|t| < \delta$, the sets A and $A + t$ both lie in an interval of length $|I| + \delta$. If $2m(A) > |I| + \delta$ they must overlap, which gives $t \in A - A$. To arrange $2m(A) > |I| + \delta$, first replace A by $A \cap J$ for an interval J where A occupies more than half of J. Such a J exists by regularity: cover A by an open set U with $m(U) < 2m(A)$ and look at its component intervals.

Hint 4. For the additive f: if $|f| \leqslant M$ on A, then $|f(a - a')| \leqslant 2M$, so f is bounded on an interval $(-\delta, \delta)$. Using $f(x/n) = f(x)/n$ this gives continuity at 0, hence everywhere. A continuous additive function is $f(x) = cx$, because it is $\mathbb{Q}$-linear and continuous.

Problem 3

Hint 1. Call a sequence good if it is monotone and satisfies the doubling condition with respect to its first term. Strengthen the statement with two parameters: if $|X| \geqslant \binom{a+b}{a} + 1$, then X contains an increasing good sequence of length $a+2$ or a decreasing good sequence of length $b+2$. The problem is the case $a = b = k - 2$. Induct on $a + b$, using Pascal's rule $\binom{a+b}{a} = \binom{a+b-1}{a-1} + \binom{a+b-1}{a}$.

Hint 2. Let $m = \min X$, $M = \max X$, $c = (m + M)/2$, and split X into $L = \{x \leqslant c\}$ and $U = \{x > c\}$. By Pascal's rule, $|L| \geqslant \binom{a-1+b}{a-1} + 1$ or $|U| \geqslant \binom{a+b-1}{a} + 1$. Apply the induction

hypothesis to that half with $(a-1,b)$ or $(a,b-1)$ respectively. The base cases $a=0$ or $b=0$ just ask for two points.

Hint 3. Key observation: an increasing good sequence $y_1 < \cdots < y_r$ inside L can be extended by M. You need $M-y_1 \geqslant 2(y_r-y_1)$, i.e. $M-y_r \geqslant y_r-y_1$, and indeed $M-y_r \geqslant M-c = c-m \geqslant y_r-y_1$. Symmetrically, a decreasing good sequence inside U can be extended by m. In each case either the half already contains the required sequence, or it contains one that is one term short in the direction that can be extended.

Problem 4

Hint 1. Divisibility: the boxes are the odd parts. Write each number as $2^s u$ with u odd. In $\{1,\ldots,2n\}$ there are only n odd values of u, and two numbers with the same odd part divide one another.

Hint 2. Monotone subsequences: the boxes are positions in an ordering. Give each term a_i the label ℓ_i, the length of the longest increasing subsequence ending at a_i. If some $\ell_i \geqslant n+1$ you are done. Otherwise n^2+1 terms have labels in $\{1,\ldots,n\}$, so $n+1$ terms share a label.

Hint 3. If $i<j$ and $\ell_i = \ell_j$, then $a_i > a_j$, since otherwise a_j would extend the sequence ending at a_i and $\ell_j > \ell_i$. Hence the $n+1$ terms with equal label form a decreasing subsequence.

Problem 5

Note. As stated ("a two-colouring ... has no monochromatic K_k") the claim is false, e.g. for the colouring with one colour. The intended statement (Erdős) is: if $n < 2^{k/2}$ (and $k \geqslant 3$), there EXISTS a two-colouring of the edges of K_n with no monochromatic K_k.

Hint 1. Count pairs (colouring, monochromatic k-set). There are $2^{\binom{n}{2}}$ colourings in total. A fixed k-set of vertices is monochromatic in exactly $2 \cdot 2^{\binom{n}{2}-\binom{k}{2}}$ of them. Equivalently: colour at random, and the expected number of monochromatic K_k is $\binom{n}{k}2^{1-\binom{k}{2}}$.

Hint 2. If this number is < 1, the bad colourings cannot cover all colourings, so a good one exists. Check: $\binom{n}{k}2^{1-\binom{k}{2}} < \frac{n^k}{k!}2^{1-k(k-1)/2} < \frac{2^{1+k/2}}{k!} \leqslant 1$ for $k \geqslant 3$, using $n^k < 2^{k^2/2}$.

Hint 3. The general principle: if the bad events B_i satisfy $\sum_i P(B_i) < 1$ (the union bound), some outcome avoids all of them. Equivalently, if (number of bad configurations) times (the largest number of objects each one spoils) is less than the total, a good object exists.